

SOHHOM BANDYOPADHYAY

AI Researcher | Cognitive Scientist | Programmer

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 <https://read.cv/sohhom>

SUMMARY

Having a blend of technical and non-technical skills, I prefer to work in areas that require data analysis, or multi-disciplinary thinking, or asking fundamental questions.

WORK EXPERIENCE

Artificial Intelligence Researcher | Playpower Labs | Remote (India)

July 2022 - April 2023 ; September 2023 - June 2024

- Reported to the CEO as an Senior Individual Contributor, occasionally collaborating with teams
- Exploring potential applications of Large Language Models and diffusion models
- Scoping and feasibility analysis for EdTech softwares e.g. for automated essay scoring
- Conducting deep and wide literature review for human preference elicitation for AI alignment

PhD Scholar | Indian Institute of Technology Gandhinagar, Center for Cognitive and Brain Sciences | Gujarat, India

July 2017 - October 2022 | Quantifying the UX of noise control headphones

- Deep dive into signals and control systems, how it relates to human sound perception
- Developed experimental protocols for measuring envelope distortion for headphones
- Conducted human experiments to measure speech perception using ANC headphones
- Full thesis PDF : <https://tinyurl.com/5n8p86jh>

Graduate Teaching Fellow | Indian Institute of Technology Gandhinagar | Gujarat, India

March - April 2019 | Co-instructor of "Computation and Cognition" (CG 501)

- Independently designed curriculum, delivered lectures and conducted exams for CG 501, a core course for graduate students in cognitive science. Taught some of the fundamentals of machine learning and game theory.

OTHER PROJECTS

Keystroke analysis: User authentication and cognitive processes from typing

MSc thesis, PhD side project (2017 - 2018)

- Applied simple neural networks to recognise humans from their typing patterns (keystrokes)
- Implemented neural network from scratch in NumPy
- Build proof-of-concept web app for multi-user authentication using keystrokes
- Explored cognitive processes from keystroke data

Natural language search interface for an academic research database

Group project as part of Natural Language Processing course (2018)

- Built a natural language query interface for a database of 500k+ research papers.
- Personal contributions: coordinated responsibilities between team members, suggested tech stack, built the data ingestion script and data schema

EDUCATION

2017 - 2022 | Indian Institute of Technology Gandhinagar

Doctor of Philosophy, Cognitive Science | CPI 8.68

2015 - 2017 | Indian Institute of Technology Gandhinagar

Master of Science, Cognitive Science | CPI 8.85

2012 - 2015 | Scottish Church College (University of Calcutta)

Bachelor of Science, Computer Science (hons) | 73.7%

TECHNICAL SKILLS

- Statistical analysis in R
- Data visualisation and EDA in Python, Matplotlib, Pandas
- Signal processing in scipy/MATLAB
- Machine learning in scikit-learn
- MVP web applications in Flask, MongoDB, Node.js and deployment in Netlify/Vercel
- Huggingface and Transformers for e.g. getting CLIP score
- Computational modelling using bespoke tools